

Phase 6 – Technical Documentation

1. Tools Used

- Python – Used for developing the automated data collection script.
- BeautifulSoup / Selenium – Used for extracting data from web pages and handling dynamic content.
- Requests Library – Used for sending HTTP requests and collecting data.
- Supabase – Used as a cloud database for storing collected information.
- VS Code – Used for development and testing.

2. Technologies Used

- Web Scraping – Extracting required information automatically from websites.
- API Integration – Collecting structured data through API responses.
- Browser Automation – Handling websites with JavaScript-based dynamic loading.
- PostgreSQL Database – Used by Supabase for structured data storage.
- Automation Workflow – Enables repeated execution without manual work.

3. Overall Architecture

Data Source



Data Collection Layer (Python + Scraping/API)



Data Processing & Cleaning



Supabase Database Storage



Data Analysis / Application Usage

4. Challenges Encountered

- Dynamic content loading using JavaScript.
- Changes in website structure.
- Difficulty in extracting structured information.
- Handling sessions, cookies, and authentication.

5. Solutions Implemented

- Used browser automation for dynamic pages.
- Monitored API calls using browser developer tools.
- Applied data cleaning and formatting techniques.
- Stored data in a structured database format using Supabase.

6. Potential Improvements

- Add real-time data monitoring.
- Deploy automation on cloud platforms.
- Improve error handling mechanisms.
- Add advanced analytics and visualization features.

Concepts Involved

- Web Scraping: Automated extraction of information from websites.
- API Collection: Fetching data through application interfaces.
- Browser Automation: Automating browser actions to collect dynamic data.
- Cookies & Sessions: Maintaining user state during web interactions.